

6/15/2026 report

Jacky Lan

Goals this quarter

- Apply oversampling techniques to dataset to increase positive sample rate for better performance
- Experiment with both MLP and XGBoosting model on if audio compression (WAV vs. FLAC) impacts model performance
- Integrating anthropogenic sounds (from BirdNet and other preprocessing models) to increase model performance

Research to read up on

- "Learning from Imbalanced Data" (He & Garcia, 2009)
 - Focus: A comprehensive overview of the entire landscape of imbalanced learning, comparing sampling techniques against algorithmic weighting and explaining the mathematical trade-offs.
- Han, Hui & Wang, Wen-Yuan & Mao, Bing-Huan. (2005). Borderline-SMOTE: A New Over-Sampling Method in Imbalanced Data Sets Learning. Adv Intell Comput. 3644. 878-887. 10.1007/11538059_91.
 - Focus: The original paper explaining why standard SMOTE fails on overlapping datasets and detailing the math behind targeting the decision boundary.
- "Data Augmentation for Environmental Sound Classification" (Salamon & Bello, 2017)
 - Focus: Highly practical. This paper details how applying pitch shifting, time stretching, and background noise mixing to raw audio significantly outperforms standard tabular data synthesis.
- "Predicting Good Probabilities With Supervised Learning" (Niculescu-Mizil & Caruana, 2005)
 - Focus: Explains why models trained on altered class distributions output distorted probabilities, and how to use Platt Scaling and Isotonic Regression to fix them.
- "A Survey of Predictive Modelling under Imbalanced Distributions" (Branco et al., 2016)
 - Focus: Discusses the shift away from standard accuracy metrics toward Precision-Recall AUC when dealing with extreme imbalances, and how different sampling techniques impact those specific metrics.